

John Holik

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Software engineer with full-stack and systems experience. Shipped a multi-service web application (CapSight AI: React, FastAPI, Node/TypeScript, PostgreSQL), deployed a live React + Cloudflare web app with a retrieval-augmented chatbot (ElectriAI), and built an end-to-end data and ML pipeline. Comfortable across the stack, from relational databases to systems-level C++ and assembly, with strong fundamentals in data structures, algorithms, operating systems, and software testing.

EDUCATION

Florida Gulf Coast University, Fort Myers, FL

May 2026

B.S. in Computer Software Engineering · Dean's List (Junior and Senior Years)

Relevant coursework: Data Structures and Algorithms, Operating Systems, Computer Security, Software Engineering, Assembly, Data Engineering, Machine Learning, Data Mining

TECHNICAL SKILLS

Languages: Python, C++, SQL, Assembly, Bash, PowerShell

Web and Backend: React, FastAPI, Node.js/Express, PostgreSQL, Tailwind CSS, Cloudflare Pages/Workers (deployed)

Testing and Tooling: Selenium, TestNG, Maven, Git/GitHub, Jupyter, Linux

ML and AI: scikit-learn, pandas, NumPy, OpenAI, Anthropic (Claude), and Gemini APIs, prompt engineering, custom agent and Claude Code skill development, LLM evaluation

Foundations: data structures and algorithms, operating systems, computer security, relational data modeling

EXPERIENCE

Undergraduate Research Assistant, then Independent Researcher (ElectriAI)

Aug 2025 – Present

U.A. Whitaker College of Engineering, Florida Gulf Coast University

- Built an end-to-end LLM pipeline analyzing an electrical-construction media corpus: data collection, GPT-based classification, human expert annotation, and rigorous evaluation.
- Processed and cleaned 330,000+ YouTube comments and ran LDA topic modeling, then redesigned the approach from unsupervised topics to a supervised, human-validated classification schema.
- Achieved 0.8965 Cohen's kappa human-vs-model agreement with per-class precision, recall, and F1 on a held-out gold set.
- Built a reproducible evaluation harness (fixed seeds, versioned data); authoring a journal manuscript (in progress).
- Built and deployed the public ElectriAI companion web app (React, Tailwind) on Cloudflare Pages with a live retrieval-augmented chatbot served through a Cloudflare Worker proxy (API-key isolation, KV-based rate limiting). Live at electriai.com.

PROJECTS

CapSight AI, LLM Pitch-Deck Grading Platform

React, FastAPI, PostgreSQL, Node/TypeScript, OpenAI

- Built a three-service application (React frontend, FastAPI/PostgreSQL persistence service, Node/TypeScript grading service) that scores uploaded decks across six categories and persists results with a history view.
- Designed REST endpoints with clean layering (routes, services, schemas) and an SQLAlchemy ORM data model; structured LLM output enforced via JSON schema.
- Hardened the service with adaptive retry, input clamping, an unreadable-input sentinel, a fallback result, and non-blocking database writes so the API stays available if the database is down.

Pro-Code AI, LLM Programming Tutor

Python, FastAPI, PostgreSQL, React, OpenAI

- Built an LLM tutor that teaches programming through guided hints and Socratic questioning rather than direct solutions.
- Designed a modular system-prompt architecture that adapts tutoring to user skill level; FastAPI backend and PostgreSQL persisted user sessions and learning history.

UFC Relational Database

PostgreSQL, SQL, pgAdmin

- Designed and implemented a normalized five-table schema modeling fighters, fights, and fight statistics with foreign-key constraints.
- Wrote SQL for insertion, filtering, and aggregation (win/loss ratios, outcomes by weight class) and tuned query performance.

Selenium + TestNG E2E Test Suite

Java, Selenium WebDriver, TestNG, Maven

- Built eight end-to-end UI test classes driving a live browser through login, posting, messaging, search, and profile flows.
- Used explicit waits and resilient locators, with ordered tests and negative-path/recovery coverage (invalid login, error handling).

Additional: Credit-Risk ML (14-model scikit-learn benchmark on 32,416 loans, 0.95 test ROC-AUC); Red Tide Reanalysis senior capstone (ensemble uncertainty quantification, reproducible Jupyter Book); Tetris recreation (C++, SFML: rendering, collision, scoring); Arduino ultrasonic security sensor (Assembly, ATmega328p).